Mathlearn · 5-sinf · Matematika

Vertikal burchaklar

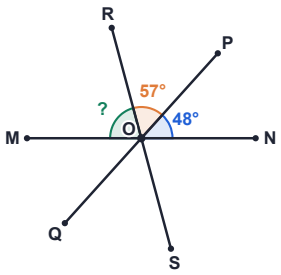
2-test (qiyinroq) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

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1. **Uch to'g'ri chiziq bir nuqtada kesishdi. Bir chiziq bo'yida yonma-yon burchaklar 48° va 57°. Shu tomondagi uchinchi burchak necha gradus?**



A) 57° **B)** 75°

C) 48° **D)** 105°

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2. **Uch chiziq kesishganda bir burchak 75°. Unga vertikal burchak necha gradus?**

A) 75° **B)** 285° **C)** 15° **D)** 105°

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3. **Ikki to'g'ri chiziq kesishganda hosil bo'lgan uchta burchak yig'indisi 290°. To'rtinchi burchak necha gradus?**

A) 70° **B)** 290° **C)** 180° **D)** 110°

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4. **Ikki vertikal burchak yig'indisi 136°. Ularning qo'shnisi necha gradus?**

A) 44° **B)** 112° **C)** 136° **D)** 68°

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5. **Kesishgan chiziqlardagi ikki qo'shni burchak 1 : 4 nisbatda. Kichigi necha gradus?**

A) 45° **B)** 36° **C)** 72° **D)** 144°

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6. **Kesishgan chiziqlardagi ikki qo'shni burchak 4 : 5 nisbatda. Kattasi necha gradus?**

A) 100° **B)** 90° **C)** 80° **D)** 125°

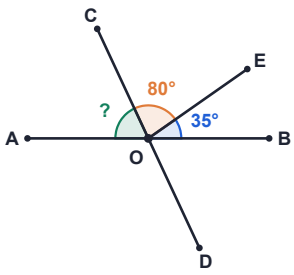
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7. **Kesishgan chiziqlarda bir burchak qo'shnisidan 40° katta. Kichigi necha gradus?**

A) 110° **B)** 70° **C)** 140° **D)** 40°

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8. **AB va CD to'g'ri chiziqlar O da kesishadi, OE — nur. ∠BOE = 35°, ∠EOC = 80°. ∠COA necha gradus?**

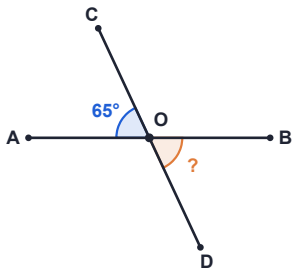


A) 35° **B)** 115°

C) 65° **D)** 80°

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9. **AB va CD to'g'ri chiziqlar O da kesishadi, ∠COA = 65°. ∠DOB necha gradus?**

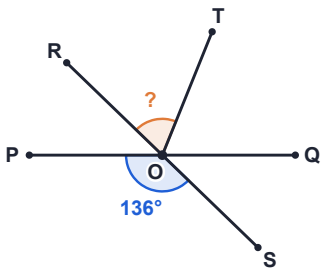


A) 65° **B)** 295°

C) 115° **D)** 25°

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10. **POQ va ROS — to'g'ri chiziqlar, ∠POS = 136°. OT nur ∠ROQ ni teng ikkiga bo'ladi. ∠ROT necha gradus?**



A) 22° **B)** 44°

C) 136° **D)** 68°

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11. **Kesishgan chiziqlardagi to'rt burchakdan biri qolgan uchtasining yig'indisidan 5 marta kichik. U necha gradus?**

A) 30° **B)** 72° **C)** 45° **D)** 60°

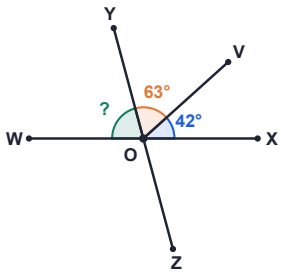
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12. **Kesishgan chiziqlarda bir burchak qo'shnisidan 3 marta katta. Kattasi necha gradus?**

A) 45° **B)** 90° **C)** 135° **D)** 120°

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13. **WOX va YOZ — to'g'ri chiziqlar, OV — nur. ∠XOV = 42°, ∠VOY = 63°. ∠YOW necha gradus?**

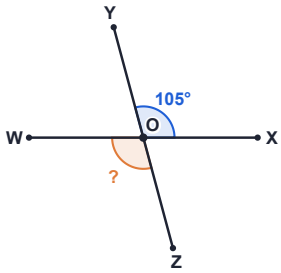


A) 42° **B)** 75°

C) 105° **D)** 63°

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14. **WOX va YOZ — to'g'ri chiziqlar, ∠XOY = 105°. ∠ZOW necha gradus?**



A) 15° **B)** 255°

C) 105° **D)** 75°

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15. **Uch to'g'ri chiziq kesishdi. Bir tomonda yonma-yon burchaklar 40° va 75°. Oltita burchakdan eng kattasi necha gradus?**

A) 75° **B)** 65° **C)** 140° **D)** 115°

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16. **Qaysi juftlik vertikal burchaklar bo'la oladi?**

A) 60° va 120° **B)** 57° va 123°

C) 90° va 180° **D)** 57° va 57°

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17. **∠a va ∠b — vertikal burchaklar, ∠a = 3x, ∠b = 72°. x necha gradus?**

A) 24° **B)** 72° **C)** 36° **D)** 216°

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18. **Vertikal burchaklardan biri 2x + 10°, ikkinchisi 3x − 20°. Bu burchak necha gradus?**

A) 110° **B)** 50° **C)** 70° **D)** 30°

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19. **Kesishgan chiziqlarda bir burchak qo'shnisidan 3 marta katta. Ikki o'tkir burchakning yig'indisi necha gradus?**

A) 180° **B)** 270° **C)** 45° **D)** 90°

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20. **Uch to'g'ri chiziq bir nuqtada kesishdi va hosil bo'lgan oltita burchak teng. Har biri necha gradus?**

A) 90° **B)** 45° **C)** 30° **D)** 60°

Javoblar

Vertikal burchaklar · 2-test (qiyinroq)

№	Javob	Qiymati
1	**B**	75°
2	**A**	75°
3	**A**	70°
4	**B**	112°
5	**B**	36°
6	**A**	100°
7	**B**	70°
8	**C**	65°
9	**A**	65°
10	**D**	68°

№	Javob	Qiymati
11	**D**	60°
12	**C**	135°
13	**B**	75°
14	**C**	105°
15	**A**	75°
16	**D**	57° va 57°
17	**A**	24°
18	**C**	70°
19	**D**	90°
20	**D**	60°

Qisqa yechimlar

1. To'g'ri chiziqdagi uch burchak yig'indisi 180°: 180° − 48° − 57° = **75°**.

2. Vertikal burchaklar teng: **75°**.

3. To'rt burchak yig'indisi 360°: 360° − 290° = **70°**.

4. 1) Har biri: 136° : 2 = 68°.
2) Qo'shnisi: 180° − 68° = **112°**.

5. 1 + 4 = 5 qism = 180°; 1 qism = **36°**.

6. 4 + 5 = 9 qism = 180°; 1 qism = 20°; kattasi 5 · 20° = **100°**.

7. (180° − 40°) : 2 = **70°** (kattasi 110°).

8. AB to'g'ri chiziq: 180° − 35° − 80° = **65°**.

9. ∠DOB — ∠COA ga vertikal: **65°**.

10. 1) ∠ROQ — ∠POS ga vertikal: 136°.
2) ∠ROT = 136° : 2 = **68°**.

11. Burchak — 1 qism, qolgani — 5 qism; jami 6 qism = 360°.
1 qism = **60°**.

12. 1 + 3 = 4 qism = 180°; 1 qism = 45°; kattasi **135°**.

13. WOX to'g'ri chiziq: 180° − 42° − 63° = **75°**.

14. ∠ZOW — ∠XOY ga vertikal: **105°**.

15. Uchinchisi: 180° − 40° − 75° = 65°. Oltita burchak: 40°, 75°, 65° va ularning vertikallari. Eng kattasi — **75°**.

16. Vertikal burchaklar teng bo'lishi shart: **57° va 57°**. (57° va 123° — qo'shni bo'la oladi.)

17. Vertikal burchaklar teng: 3x = 72°, x = **24°**.

18. 1) Teng: 2x + 10° = 3x − 20°, x = 30°.
2) Burchak: 2 · 30° + 10° = **70°**.

19. Kichigi 45°, kattasi 135°. O'tkir burchaklar — ikki vertikal 45°: 45° + 45° = **90°**.

20. Oltita burchak yig'indisi 360°: 360° : 6 = **60°**.